# A Complete Characterization of Property-$(R_1)$ Kernels in Finite-Dimensional $\ell_\infty$

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August 9, 2026

## Result Status

**Candidate full solution, likely valid, pending expert review.** The result answers the characterization problem in Remark 5.2(a) of Das–Paul [1]. The answer depends only on the row space of the given functionals: after invertible row operations, their matrix must be a signed coordinate-incidence matrix.

## 1 Source Problem

The source is Syamantak Das and Tanmoy Paul, *On property-$(R_1)$ and relative Chebyshev centers in Banach spaces-II*, arXiv:2207.09623 [1]. The question appears in Remark 5.2(a), page 17 of the arXiv PDF.

> It is clear that $y = (y(n)) \in \bigoplus_{c_0} Y$, because $\mathrm{rad}_{B_Y}(F(n)) \to 0$.
>
> Since for any $z = (z(1), z(2), \cdots) \in \bigoplus_{c_0} B_Y$, $r(y, F) \leq r(z, F)$ and because $\|(y(n))\|_\infty \leq 1$, $y \in \mathrm{Cent}_{\bigoplus_{c_0} B_Y}(F) = \mathrm{Cent}_{B_{Y_0}}(F)$, the result follows from Theorem 1.4. $\square$
>
> **Remark 5.2.** (*a*) *If $Y$ is a finite co-dimensional proximinal subspace of $c_0$, then there exists $n \in \mathbb{N}$, such that $Y = F \oplus_\infty Z$, where $F$ is a subspace of $\ell_\infty(n)$ and $Z = \{(x_i) \in c_0 : x_i = 0, 1 \leq i \leq n\}$. From Theorem 4.4, it is clear that $Y$ has property-$(R_1)$ in $c_0$ if and only if $F$ has property-$(R_1)$ in $\ell_\infty(n)$. We do not know the characterization of $\alpha_i \in \ell_1(n)$ where $1 \leq i \leq m$, considering $\dim(c_0/Y) = m$, satisfying the condition that $\cap_i \ker(\alpha_i)$ has property-$(R_1)$ in $\ell_\infty(n)$.*
> (*b*) *If $Y$ is a finite co-dimensional proximinal subspace of $c_0$, then from the decomposition $Y = F \oplus_\infty Z$, it is also clear that $Y$ has property-$(R_1)$ in $c_0$ if and only if $Y$ has property-$(R_1)$ in $\ell_\infty$.*

In the finite-dimensional part of that remark, the problem is to characterize families

$$\alpha_1, \ldots, \alpha_m \in \ell_1^n$$

satisfying

$$\dim\left(\ell_\infty^n \Big/ \bigcap_i \ker \alpha_i\right) = m,$$

for which

$$V := \bigcap_{i=1}^m \ker \alpha_i$$

has property $(R_1)$ in $\ell_\infty^n$, with respect to all finite subsets.

## 2 Definitions

For a nonempty finite set $E \subset \ell_\infty^n$, put

$$S_s(E) := \bigcap_{x \in E} B_\infty(x, s), \qquad r(v, E) := \max_{x \in E} \|v - x\|_\infty .$$

The triplet $(\ell_\infty^n, V, \mathcal{F}(\ell_\infty^n))$ has *strong property* $(R_1)$ if, whenever $v \in V$, $r_1, r_2 > 0$, and $E$ is finite,

$$r(v, E) \le r_1 + r_2, \qquad S_{r_2}(E) \cap V \neq \varnothing,$$

one has

$$V \cap B_\infty(v, r_1) \cap S_{r_2}(E) \neq \varnothing.$$

Ordinary property $(R_1)$ is the corresponding approximate/open-radius condition stated in the source paper.

For $r \ge 0$, define coordinatewise clipping by

$$(T_r x)_j := \operatorname{sgn}(x_j) \min\{|x_j|, r\}.$$

We identify $(\ell_\infty^n)^* = \ell_1^n$ with $\mathbb{R}^n$, using the standard coordinate vectors $e_1, \ldots, e_n$.

## 3 Proof Intuition

Intersections of equal-radius $\ell_\infty$-balls are precisely bounded axis-parallel boxes. Property $(R_1)$ says that if such a box meets the subspace and also meets a cube centered at a point of the subspace, then all three sets meet. Centering at zero turns this into a nearest-point statement for boxes. The closest point of a box to zero is obtained by clipping any point in the box toward the coordinate cube. Thus the entire metric condition is equivalent to the algebraic rule $T_r(V) \subset V$.

A truncation-stable linear subspace is rigid. A nonzero vector with smallest support clips to a sign vector on that support. If another vector were not a scalar multiple of that sign pattern on the same support, clipping a small perturbation would create a nonzero vector with strictly smaller support. Removing the resulting signed diagonal block and iterating gives the full decomposition. Taking annihilators converts the decomposition into the promised condition on the $\alpha_i$.

## 4 Main Theorem

**Theorem 1** (Complete characterization)**.** *Let $\alpha_1, \ldots, \alpha_m \in \ell_1^n$ be linearly independent, let $A$ be the $m \times n$ matrix with these vectors as rows, and let*

$$V = \ker A = \bigcap_{i=1}^{m} \ker \alpha_i \subset \ell_\infty^n.$$

*The following are equivalent.*

1. *$(\ell_\infty^n, V, \mathcal{F}(\ell_\infty^n))$ has property $(R_1)$.*
2. *The same triplet has strong property $(R_1)$.*
3. *$T_r(V) \subset V$ for every $r \ge 0$.*

4. *After a permutation of coordinates, there are pairwise disjoint blocks $B_1, \dots, B_q$, sign vectors $s^{(a)} \in \{\pm 1\}^{B_a}$, a set $Z$ of zero coordinates, and a set $J$ of free singleton coordinates such that*

$$V = \{x \in \mathbb{R}^n : x_j = 0 \ (j \in Z), \quad x|_{B_a} = t_a s^{(a)} \ (t_a \in \mathbb{R}), \quad x_j \textit{ arbitrary } (j \in J)\}.$$

5. *The row space* $\mathrm{row}(A) = V^\perp$ *admits a basis whose members, after individual nonzero rescaling, have one of the forms*

$$e_j, \qquad e_j + e_k, \qquad e_j - e_k \quad (j \neq k).$$

*Consequently, condition (5) is the requested characterization of the family $\alpha_1, \dots, \alpha_m$. Equivalently, invertible row operations turn $A$ into a matrix whose rows have either one nonzero entry or two entries of equal absolute value.*

The proof is divided into three elementary lemmas.

## 5 From Property $(R_1)$ to Boxes

**Lemma 1.** *For a linear subspace $V \subset \ell_\infty^n$, property $(R_1)$ for all finite subsets is equivalent to strong property $(R_1)$.*

*Proof.* The closed unit ball $B_V$ is compact. For every finite $E$, the function $y \mapsto r(y, E)$ is continuous, so it attains its minimum on $B_V$. Thus $\mathrm{Cent}_{B_V}(E) \neq \varnothing$. The equivalence stated in [1, Theorem 1.4] says that strong property $(R_1)$ is equivalent to ordinary property $(R_1)$ together with this center-existence condition. The conclusion follows. □

Call a product $Q = \prod_{j=1}^n [a_j, b_j]$, with $a_j \le b_j$, a bounded coordinate box.

**Lemma 2** (Box reformulation)**.** *Strong property $(R_1)$ is equivalent to the following condition: for every bounded coordinate box $Q$, every $v \in V$, and every $r > 0$,*

$$Q \cap V \neq \varnothing, \quad Q \cap B_\infty(v, r) \neq \varnothing \quad \Longrightarrow \quad Q \cap V \cap B_\infty(v, r) \neq \varnothing. \tag{B}$$

*Proof.* First note that every $S_s(E)$ is a bounded coordinate box. Conversely, let $Q = \prod_j [a_j, b_j]$. Choose

$$s > 0, \qquad 2s \ge \max_j (b_j - a_j), \qquad c_j = \frac{a_j + b_j}{2}.$$

For each $j$, form two points $p^{j,+}, p^{j,-}$ by taking midpoint coordinates except at $j$, and setting

$$p_j^{j,+} = a_j + s, \qquad p_j^{j,-} = b_j - s.$$

For $E = \{p^{j,+}, p^{j,-} : 1 \le j \le n\}$, direct coordinatewise calculation gives $S_s(E) = Q$.

Suppose strong property $(R_1)$ holds, and let $z \in Q \cap B_\infty(v, r)$. With the representation $Q = S_s(E)$, for every $x \in E$ we have

$$\|v - x\|_\infty \le \|v - z\|_\infty + \|z - x\|_\infty \le r + s.$$

Together with $Q \cap V \neq \varnothing$, strong property $(R_1)$ gives (B).

Conversely, suppose (B) holds and the hypotheses of strong property $(R_1)$ are satisfied. Put $Q = S_{r_2}(E)$. In coordinate $j$,

$$Q_j = \left[ \max_{x \in E} (x_j - r_2), \ \min_{x \in E} (x_j + r_2) \right].$$

The inequalities $\|v - x\|_\infty \le r_1 + r_2$, for all $x \in E$, say exactly that $Q_j$ intersects $[v_j - r_1, v_j + r_1]$ for every $j$. Hence $Q \cap B_\infty(v, r_1) \neq \varnothing$. Applying (B) proves strong property $(R_1)$. □

## 6 Boxes Are Coordinatewise Clipping

**Lemma 3** (Clipping criterion)**.** *Condition* (B) *is equivalent to $T_r(V) \subset V$ for every $r \geq 0$.*

*Proof.* Because $V$ is linear, translating $Q$ by $-v$ shows that it is enough to use (B) with $v = 0$.

Assume first that $T_r(V) \subset V$. Let $Q \cap V \neq \varnothing$, choose $x \in Q \cap V$, and suppose $Q \cap B_\infty(0, r) \neq \varnothing$. For each coordinate interval $[a_j, b_j]$ of $Q$, the latter condition says $[a_j, b_j] \cap [-r, r] \neq \varnothing$. Moving $x_j$ toward $[-r, r]$ therefore stays inside $[a_j, b_j]$, so $T_r x \in Q$. By assumption $T_r x \in V$, and by definition $T_r x \in B_\infty(0, r)$. Thus (B) holds.

Conversely, fix $x \in V$ and $r \geq 0$. The cases $r = 0$ and $r \geq \|x\|_\infty$ are immediate. Otherwise define the box $Q(x, r) = \prod_j I_j$, where

$$I_j = \begin{cases} [r, x_j], & x_j > r, \\ [x_j, -r], & x_j < -r, \\ \{x_j\}, & |x_j| \leq r. \end{cases}$$

It contains $x \in V$, while

$$Q(x, r) \cap B_\infty(0, r) = \{T_r x\}.$$

Condition (B) forces this unique point into $V$. Hence $T_r(V) \subset V$. □

## 7 Classification of Clipping-Stable Subspaces

**Lemma 4** (Minimal-support decomposition)**.** *A linear subspace $V \subset \mathbb{R}^n$ is invariant under every $T_r$ if and only if it has the signed-block decomposition in Theorem 1(4).*

*Proof.* The stated block spaces are visibly clipping-stable: clipping a signed block $ts$ gives $\operatorname{sgn}(t) \min\{|t|, r\} s$, and clipping preserves zero and singleton blocks.

For the converse, suppose $V \neq \{0\}$, and choose $x \in V \setminus \{0\}$ for which $|\operatorname{supp} x|$ is minimal. If $0 < r < \min_{j \in \operatorname{supp} x} |x_j|$, then

$$s := r^{-1} T_r x \in V$$

is a sign vector supported on the same set $B := \operatorname{supp} x$.

We claim that for every $y \in V$ there is a scalar $c(y)$ such that

$$y_j = c(y) s_j \qquad (j \in B). \tag{1}$$

Indeed, suppose the numbers $s_j y_j$, $j \in B$, are not all equal. Choose $t > 0$ so small that every coordinate of $s + ty$ on $B$ keeps the sign $s_j$, and every coordinate off $B$ has absolute value smaller than every coordinate on $B$. On $B$,

$$|s_j + t y_j| = 1 + t s_j y_j.$$

Choose a clipping level $\rho$ strictly between two distinct values in this finite list, avoiding all of them. Then

$$w := T_\rho(s + ty) - (s + ty) \in V$$

is nonzero, has no support off $B$, and is supported on the nonempty proper subset of indices where $|s_j + t y_j| > \rho$. This contradicts the minimality of $B$, proving (1).

It follows that

$$V = \operatorname{span}\{s\} \oplus V_0, \qquad V_0 := \{y \in V : y|_B = 0\}.$$

The second summand is again invariant under all coordinatewise clippings and is supported off $B$. Iteration terminates after finitely many steps and gives disjoint signed diagonal blocks. Coordinates never appearing in a support are zero on $V$; blocks of size one are precisely free coordinate directions. This is the required decomposition. □

*Proof of Theorem 1.* Lemmas 1, 2, 3, and 4 prove the equivalence of (1)–(4).

It remains to translate (4) into (5). A zero coordinate $j \in Z$ is cut out by the normal $e_j$. For each signed block $B_a$, choose a base index $j_a \in B_a$. The block condition $x|_{B_a} = t_a s^{(a)}$ is cut out by the independent normals

$$s^{(a)}_{j_a} e_{j_a} - s^{(a)}_j e_j, \qquad j \in B_a \setminus \{j_a\}.$$

Thus $V^\perp$ has a basis of the form in (5).

Conversely, suppose $V^\perp$ is spanned by normals $e_j$ and $e_j \pm e_k$. Regard every two-coordinate normal as a signed edge between vertices $j, k$. In each connected component the equations determine every coordinate as a fixed sign times one scalar, unless a one-coordinate equation or an inconsistent signed cycle pins the entire component to zero. Isolated unconstrained vertices are free singleton blocks. Hence $V$ has the decomposition in (4).

Finally, $\mathrm{row}(A) = V^\perp$, and its dimension is $m$. A spanning family of elementary normals therefore contains an elementary-normal basis, while any such basis is obtained from the original rows of $A$ by invertible row operations. This proves (5) and the final formulation. □

**Corollary 1** (Concrete obstruction)**.** *The line $V = \ker(1, 2) = \mathrm{span}\{(2, -1)\} \subset \ell^2_\infty$ does not have property $(R_1)$.*

*Proof.* For $x = (2, -1) \in V$, one has $T_1 x = (1, -1) \notin V$. Equivalently, the box $Q = [1, 2] \times \{-1\}$ meets $V$ at $x$, but its unique point in the unit ball is $(1, -1) \notin V$. Lemma 3 applies. □

# 8 Relation to Earlier Literature

Miesch–Pavón [2, Theorem 4.7] characterize weakly externally hyperconvex linear subspaces of $\ell^n_\infty$. Their answer is the same signed-coordinate class as Theorem 1(4)–(5): precisely the intersections of hyperplanes with normals $e_j$ or $e_j \pm e_k$. Their paper does not discuss property $(R_1)$, and their theorem is not used in the proof above. The coincidence supplies an independent structural cross-check; the box/clipping equivalence identifies property $(R_1)$ with that previously occurring class in finite-dimensional $\ell_\infty$.

# 9 Verification Notes

The proof was audited around four possible failure points.

1. Ordinary property becomes strong property because $B_V$ is compact; this is exactly the extra hypothesis in [1, Theorem 1.4].

2. The common-radius representation really includes every bounded box, including degenerate intervals, because $b_j - s \leq a_j + s$.

3. In Lemma 4, $t$ is chosen before $\rho$, so no coordinate outside the minimal support is clipped. The clipping difference is therefore supported on a strict nonempty subset of that support.

4. The criterion must concern $\mathrm{row}(A)$, not each originally presented $\alpha_i$. Property $(R_1)$ depends only on $\ker A$, and invertible row operations do not change that kernel.

No numerical or computer-assisted step is used in the proof. The example $\ker(1, 2)$ is a direct sanity check showing how a forbidden coefficient ratio produces a failed clipping.

## 10 Novelty Check and Limitations

Before promotion, the run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2207.09623, the paper title, property $(R_1)$, and the finite-dimensional $\ell_\infty$ kernel problem; no duplicate result was found. Targeted web searches on August 9, 2026 used the exact source sentence and close combinations of "property $(R_1)$", "finite dimensional", "$\ell_\infty$", "kernel", and "characterization". They found the source paper and general Chebyshev-center literature, but no later paper stating this characterization.

The closest primary result found was the earlier Miesch–Pavón theorem just discussed. Because it concerns weak external hyperconvexity rather than property $(R_1)$, it is recorded as related literature rather than an existing answer. Novelty confidence is therefore moderate, not definitive: an expert should check older terminology around the $1\frac{1}{2}$-ball property and ball-intersection subspaces for an implicit equivalent formulation.

The theorem is real and finite-dimensional, exactly as in the source remark. No claim is made here for complex scalars or arbitrary subspaces of infinite $\ell_\infty$.

## 11 Human Review Recommendation

Send for expert review as a candidate full solution. The most useful review is to verify the equivalence between the source's ordinary and strong definitions, then inspect the minimal-support clipping argument. A secondary literature reviewer should decide whether the new box/clipping identification is already implicit in the older $1\frac{1}{2}$-ball literature; this affects novelty classification but not the self-contained proof.

## References


[1] S. Das and T. Paul, *On property-$(R_1)$ and relative Chebyshev centers in Banach spaces-II*, arXiv:2207.09623; Quaestiones Mathematicae **47** (2024), 461–476.

[2] B. Miesch and M. Pavón, *Weakly externally hyperconvex subsets and hyperconvex gluings*, arXiv:1507.07795, Theorem 4.7.